Stochastic Attention via Langevin Dynamics on the Modern Hopfield Energy

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Motivation

Attention heads are deterministic: the same query always returns the same softmax-weighted memory average. Yet many tasks require *generating* novel, plausible patterns from structured memory. Score-based diffusion models solve this for large datasets by learning a neural score function, but they require thousands of training examples and expensive optimization. When data is scarce, there is nothing to train on. We've explored a different approach: instead of learning a score, we can **derive** the exact score function from the modern Hopfield energy landscape. We observe that the gradient of the **modern Hopfield energy** is exactly the identity minus the attention map. Applying **Langevin dynamics** to this energy yields *stochastic attention*: a **training-free sampler** controlled by a single inverse temperature β (Figure 1).

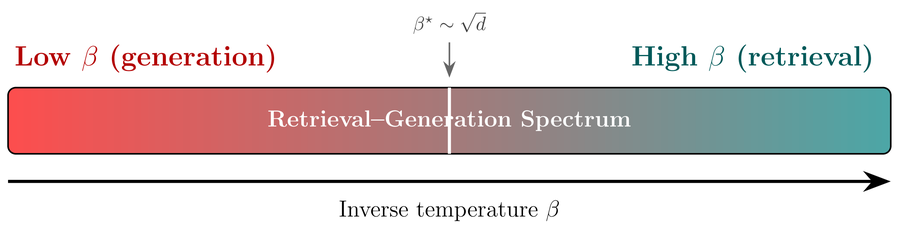


Figure 1. The retrieval–generation spectrum. A single inverse temperature β controls sample behavior: low β yields broad generation, high β yields exact retrieval. The critical temperature $\beta^* \sim \sqrt{d}$ marks the boundary, identified analytically via an entropy inflection condition for random unit-norm memories. A single parameter β interpolates continuously between open-ended generation ($\beta \to 0$) and exact retrieval ($\beta \to \infty$).

Method: Stochastic Attention Update

Suppose you have a memory (pattern) matrix $\mathbf{X} = [\mathbf{m}_1, \ldots, \mathbf{m}_K] \in \mathbb{R}^{d \times K}$ that stores K patterns where each each pattern is given by $\mathbf{m}_k \in \mathbb{R}^d$. To generate a new pattern or retrieve an existing one, we sample from $p_\beta(\boldsymbol{\xi}) \propto \exp(-\beta E(\boldsymbol{\xi}))$. The Unadjusted Langevin Algorithm (ULA) turns any smooth energy U into a sampler:

$$\boldsymbol{\xi}_{t+1} = \boldsymbol{\xi}_t - \tilde{\alpha}\, \nabla U(\boldsymbol{\xi}_t) + \sqrt{2\tilde{\alpha}}\, \boldsymbol{\epsilon}_t, \qquad \boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{I}).$$

The gradient of the modern Hopfield energy is exactly the attention residual:

$$\nabla E(\boldsymbol{\xi}) = \boldsymbol{\xi} - \underbrace{\mathbf{X}\,\mathrm{softmax}(\beta\, \mathbf{X}^\top \boldsymbol{\xi})}_{\mathbf{T}(\boldsymbol{\xi}) \,=\, \text{softmax attention map}} .$$

Substituting ∇E into the ULA (with $U = \beta E$, $\tilde{\alpha} = \alpha/\beta$):

$$\boldsymbol{\xi}_{t+1} = \underbrace{(1-\alpha)\, \boldsymbol{\xi}_t}_{\text{contraction}} + \underbrace{\alpha\, \mathbf{X}\,\mathrm{softmax}(\beta\, \mathbf{X}^\top \boldsymbol{\xi}_t)}_{\text{attention pull to memories}} + \underbrace{\sqrt{\tfrac{2\alpha}{\beta}}\, \boldsymbol{\epsilon}_t}_{\text{Gaussian noise}}$$

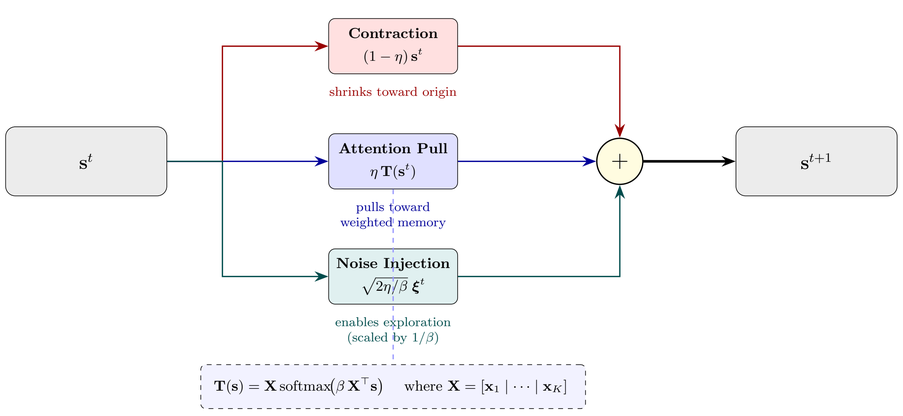


Figure 2. The stochastic attention update decomposes into three components: contraction toward the origin, an attention pull toward stored memories, and isotropic Gaussian noise scaled by temperature. Each step costs $\mathcal{O}(dK)$, with no training required. Convergence is W_2-geometric in the convex regime ($\beta\sigma_{\max}^2 < 2$); Metropolis-Adjusted Langevin Algorithm (MALA) acceptance = 99.2% at $\alpha = 0.01$ confirms negligible discretization bias. The critical temperature β^* is selected via the entropy inflection condition and scales as $\beta^* \sim \sqrt{d}$.

Experiment 1: Phase Behavior & Convergence (Synthetic)

The theory predicts that β acts as a continuous order parameter, with a smooth transition between generation and retrieval centered near $\beta^* \sim \sqrt{d}$. We tested this on synthetic random unit-norm patterns and verified convergence to the correct Boltzmann target (Figure 3).

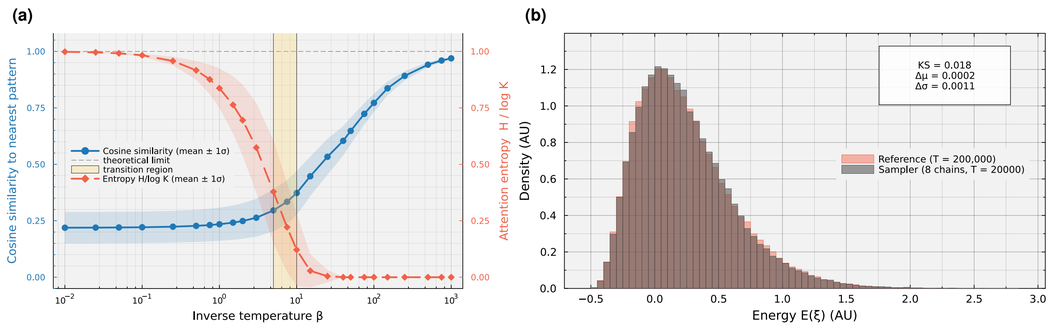


Figure 3. Synthetic data experiments. **(a)** $d = 64$, $K = 16$. Both cosine similarity to the nearest pattern (blue) and attention entropy (coral) underwent a smooth sigmoidal transition centered at $\beta \approx 5$–10 (gold band), consistent with the predicted $\beta^* \sim \sqrt{d} \approx 8$. **(b)** $d = 8$, $K = 4$, $\beta = 5$. Eight independent chains (gray) converged to a long-run reference distribution (coral); Kolmogorov–Smirnov (KS) testing confirmed the sampler reached the correct Boltzmann target.

References

[1] Ramsauer et al., "Hopfield Networks is All You Need," *ICLR*, 2021.
[2] Roberts & Tweedie, "Exponential Convergence of Langevin Distributions and Their Discrete Approximations," *Bernoulli*, 2(4):341–363, 1996.
[3] Ho, Jain & Abbeel, "Denoising Diffusion Probabilistic Models," *NeurIPS*, 33:6840–6851, 2020.
[4] Durmus & Moulines, "Nonasymptotic Convergence Analysis for the Unadjusted Langevin Algorithm," *Ann. Appl. Probab.*, 27(3):1551–1587, 2017.

Experiment 2: MNIST Image Generation ($K = 100$, $d = 784$)

With the phase behavior confirmed, we moved to structured data. We stored $K = 100$ images of digit "3" and compared stochastic attention against both synthetic baselines (bootstrap, Gaussian perturbation, convex combinations, Gaussian Mixture Model with Principal Component Analysis (GMM-PCA)) and learned generative models (Variational Autoencoder (VAE), Denoising Diffusion Probabilistic Model (DDPM)). We ran 30 independent chains at $\beta = 2000$ (retrieval) and $\beta = 200$ (generation, near β^*), collecting 150 samples per regime (Figure 4).

Method	Nov.↑	Div.↑	Energy↓
Bootstrap	0.000	0.459	−0.500
Gaussian perturb.	0.004	0.450	−0.496
Convex combo	0.092	0.008	−0.399
GMM-PCA	0.198	0.419	−0.303
VAE (latent = 8)	0.214	0.441	−0.286
DDPM†	0.938	0.991	+0.44
MALA ($\beta = 2000$)	0.151	0.598	−0.305
SA ($\beta = 2000$)	0.152	0.600	−0.303
SA ($\beta = 200$)	**0.548**	**0.885**	+1.467

†DDPM generated isotropic noise despite 5,000 training epochs.

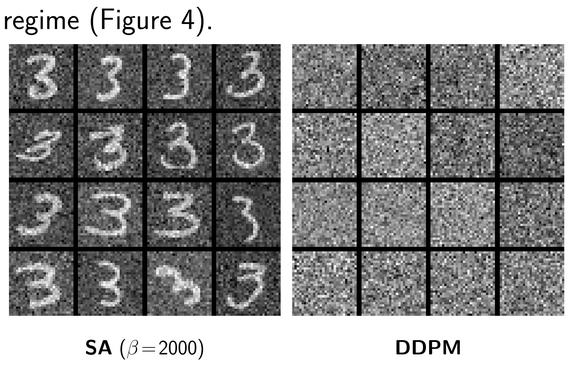
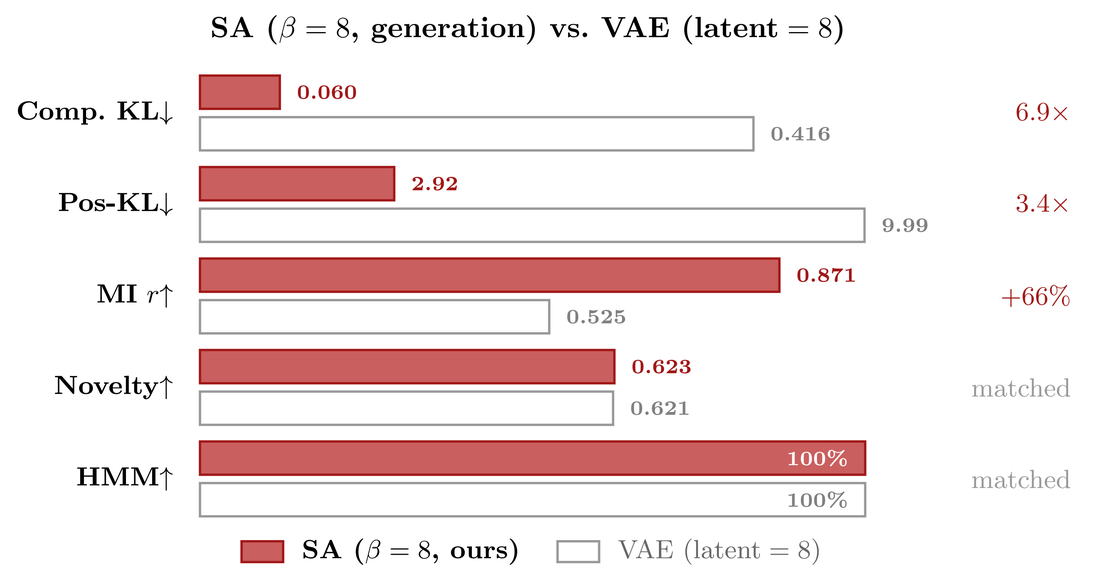


Figure 4. (Left) Novelty, diversity, and energy scores for all methods on MNIST digit "3" ($K = 100$, $d = 784$). SA at generation temperature ($\beta = 200$) achieved the highest novelty and diversity among energy-based methods. **(Right)** Sample grids from SA ($\beta = 2000$, retrieval regime) and DDPM (5,000 training epochs). DDPM produced isotropic noise on this small dataset.

In the generation regime, SA achieved **2.6**× the novelty and **2.0**× the diversity of the best learned baseline (VAE), while matching the Metropolis-corrected MALA gold standard. DDPM failed entirely on this small dataset, producing isotropic noise despite 5,000 training epochs. This illustrates the core advantage of a training-free score: when data is scarce, there is no distribution to learn, but the exact Hopfield gradient is always available.

Experiment 3: Protein Sequence Generation (Pfam RRM, $K = 68$, $d = 59$)

Protein sequences posed a harder test than images: generated samples must preserve not just global amino acid composition but position-specific conservation and pairwise co-evolutionary couplings that define family membership. We stored 68 aligned RRM (RNA Recognition Motif) sequences from Pfam, one-hot encoded and PCA-reduced to $d = 59$ (95% variance retained). The entropy inflection gave $\beta^* = 3.85$, half the random-pattern prediction $\sqrt{d} = 7.68$, because conserved residues concentrate the similarity spectrum.



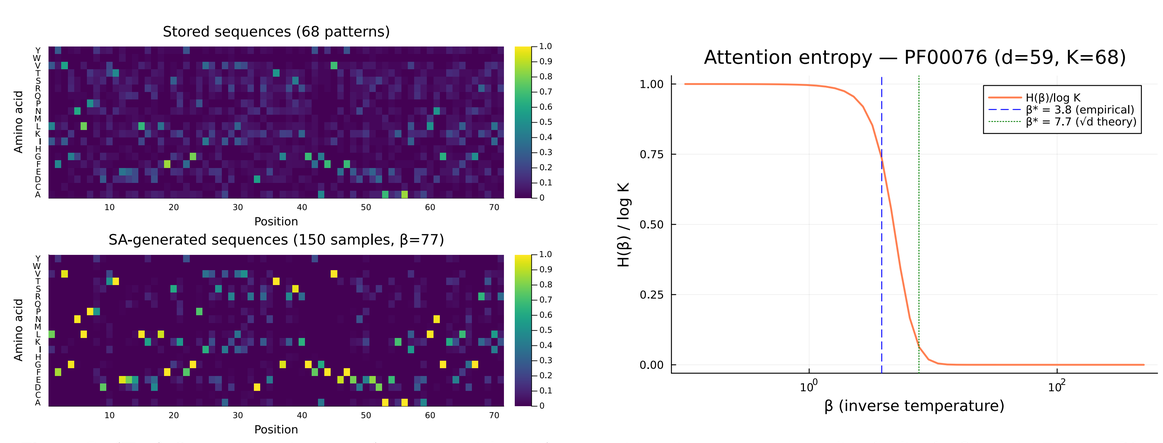


Figure 5. (Top) Composition divergence (Kullback–Leibler, KL), per-position divergence, and coupling fidelity for SA vs. VAE on the Pfam RRM family ($K = 68$, $d = 59$). **(Bottom-left)** Stored (top) vs. SA-generated (bottom) amino acid frequencies. **(Bottom-right)** Entropy inflection identified $\beta^* = 3.85$, the retrieval–generation boundary.

At generation temperature, SA matched the VAE on novelty while achieving **6.9**× lower composition divergence, **3.4**× lower per-position divergence, and **66**% higher coupling fidelity. Every generated sequence passed the Pfam Hidden Markov Model (HMM) filter (median E-value 8.1×10^{-34}). Because the Langevin gradient was recomputed from stored sequences at every step, it continuously steered samples toward the family's structural profile – fidelity that the VAE, trained on only 68 examples, could not maintain.

Key Result

Stochastic attention is a **training-free** generative sampler whose score function is the attention map itself. Each step costs $\mathcal{O}(dK)$ – the same as one attention head – with no learned parameters, no training loop, and no dataset size requirement. A single analytically identified parameter β^* controls the generation–retrieval tradeoff.

On **MNIST** ($K = 100$), SA achieved **2.6**× the novelty and **2.0**× the diversity of the best learned baseline (VAE), while DDPM failed entirely. On **protein sequences** ($K = 68$), SA achieved **6.9**× lower composition divergence and **66**% higher coupling fidelity than the VAE, with 100% Pfam HMM pass rate.